# Supplementary Information

WestQuant Memory Discovery Lab: An Evidence-Aware Framework from Experimental Data to Quantum-Memory Simulation

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This Supplement describes the executed software and data analyses accompanying the manuscript. Experimental reanalysis, internal diagnostics, synthetic controls and numerical verification are identified separately. No new hardware experiment, independently verified new-material lifetime, or machine-learning training result is reported. Tables are generated from the archived JSON outputs where applicable.

## S1 Executable interfaces and conventions

The package `memorylab` provides candidate/source records, a catalog, memory profiles, model dispatch, quantum channels, deterministic $T_2$ assessment and network export. Source identities are content hashes with source metadata. The base evaluated source is version 0.6.0; its archive identity is recorded in the reproduction section. The additional system evaluation package is version 0.1.0 and retains a separate source fingerprint.

| Component | Input contract and interpretation |
|---|---|
| Candidate | Composition, integer charge, electronic spin $2S$, supplied geometry, optional host and site-resolved isotope substitutions. Electron/spin parity is checked; stability is not established. |
| Observation | Quantity, unit, conditions, source location and evidence kind. Unknown units and unresolved observable definitions remain unresolved. |
| Reviewed reference | Material identity, coherence definition, positive seconds, reported uncertainty convention, conditions and unresolved conditions. Matching is conditional on reported information; no reference-query interpolation. |
| Memory profile | Family, logical states, writing/storage/reading temperatures and durations, explicit relaxation/dephasing definitions, IO acceptance and optional independently specified bath/table coherence. |
| Exact bath | Nuclear $I$, Zeeman vectors and coupling tensors in Hz; dimension bounded; time in seconds; ideal pulse sequence; bath Gibbs initialization at specified temperature. |
| PyCCE | Conventional CCE pure coherence; backend milliseconds, kHz and gauss; explicit conversion from the common SI interface. Density elements from generalized CCE are not silently substituted. |
| Electronic adapters | Molecular PySCF geometry in angstrom, charge and $2S$; external coupling imports require units, site mapping and tensor-frame alignment. Periodic crystals are not molecular PySCF inputs. |
| Optical propagation | Separate MaxwellBloch multi-level recipe. Propagation fields do not by themselves identify a single-photon instrument. |
| Channel | Conditional logical CPTP map and state-independent acceptance; full qubit-to-qutrit erasure map; unnormalized column-vectorized Choi matrix. |
| Network device | Explicit age samples, exact requested temperature/protocol, retrieval from original input once. Missing age or unsupported intervention is an error. |

Table S1: Common interface conventions and inference boundaries.

The implemented family labels are atomic ensemble, trapped atom/ion, rare-earth solid, defect spin, molecular spin and negative ion. Their availability is not six independent experimental validations. The negative-ion

branch requires an environment-overlap parameter; it does not deduce coherence from neutralization. Temperature ramps, state-dependent acceptance and a complete structure-to-spin–phonon-to-optical-instrument chain are outside the evaluated contract.

Channel compression retains Choi eigenvalues above $10^{-14}$ and rejects eigenvalues below $-10^{-9}$. Default CPTP checks use $10^{-9}$ tolerance; the QuTiP Markov comparison permits $2 \times 10^{-8}$. State-validity checks use $10^{-9}$. Recorded maximum discrepancies, rather than merely boolean checks, are retained. Numerical agreement is verification of the specified equations, not evidence that omitted experimental physics is negligible.

## S2 Source audit and reviewed coherence values

The main experimental sources are Zenodo records 14929997 and 21532904, associated with papers 10.1103/z6lc-qw2d and 10.1103/ftkb-pkvp. Both data records declare CC BY 4.0. Fourteen Teller artifacts are recorded, including processed figures and reference-source snapshots. The deterministic data branch has eleven artifacts: the PR record metadata and seven measurement files, plus three pinned molecular-repository artifacts. All seven PR measurement-file MD5 values are checked against the record metadata, and each cached artifact also has a SHA-256 digest.

The molecular data are pinned to the MTD-group molecular-qubit repository revision

`5263b1d86b39243e7405dde5bf7c58bbd8fcd677`.

Positive entries are provisional because a uniform unit, pulse definition and operating conditions have not been reviewed. There are 201 such entries across 123 source rows and 40 paper links. The repository license is not specified in the retrieved source metadata. These cached files are present for the local reproducibility bundle and excluded from default Git tracking. Their presence does not confer a license for public redistribution.

| Material / source | Observable / protocol | Reported time |
|---|---|---|
| $^{151}Eu^{3+}$:YSO; Ortu et al. | Spin DD, XX | $70 \pm 2$ ms |
| Same | Spin DD, XY4 | $106 \pm 9$ ms |
| Same | Spin DD, XY8 | $154 \pm 11$ ms |
| Same | Spin DD, XY16 | $230 \pm 30$ ms |
| Same | Effective optical AFC | $240 \pm 30$ $\mu$s |
| V(IV) complex 1; Lin et al. | Hahn stretched $T_m$ | $2.04 \pm 0.05$ $\mu$s |
| V(IV) complex 2; Lin et al. | Hahn stretched $T_m$ | $4.11 \pm 0.03$ $\mu$s |
| V(IV) complex 3; Lin et al. | Hahn stretched $T_m$ | $4.42 \pm 0.02$ $\mu$s |
| PL6 in 4H-SiCOI; Hu et al. | Hahn $T_2$ | $4.62 \pm 1.65$ $\mu$s |
| Same | Ramsey $T_2^*$ | $460 \pm 35$ ns |

Table S2: Ten reviewed primary-text values. These are different definitions, not a common material ranking, and are not fresh fits to raw trials. The Eu intervals are reported 95% confidence intervals. The uncertainty conventions for the other studies have not been verified.

The Eu references use an approximately 4 K cryostat and 1.35 mT along $D_1$. Local RF heating, the detailed bath and sequence-specific stretch exponents remain unresolved. The optical AFC expression has exponent factor four, distinct from the spin efficiency expression. See Ortu et al., npj Quantum Information 8, 29 (2022).

The molecular references apply at 5 K in an o-terphenyl glass at X-band, approximately 9.4 GHz. Complexes 1–3 have $Et_3NH$, $n$-$Bu_3NH$ and $n$-$Hex_3NH$ counterions, respectively, with $[V(C_6H_4O_2)_3]$ core. Exact selected fields, site isotope distributions and fully reviewed concentrations are unavailable in the current reference table. See Lin et al., Chemical Science 10, 548–555 (2019). The PL6 references apply to a natural-abundance 4H-SiC-on-insulator waveguide at reported room temperature; a numeric temperature and exact coherence-measurement field remain missing. See Hu et al., Nature Communications 15, 10256 (2024). None of these ten records validates an entire optical quantum-memory channel.

## S3 Teller data reanalysis and internal diagnostics

The parser handles source indices explicitly. For `4.csv`, rows encode basis and zero-based cell; the six labels are X, −X, Y, −Y, 0, 1. For `8a.csv`, the basis precedes a one-based cell index; its labels are X, −X, Y, −Y,

E, L. Original headers alone would give an incorrect interpretation. Uniform means of the processed state fidelities reproduce the released figure summaries. These numerical identity checks use neither inferred raw counts nor independent experimental replication.

The spatial holdout uses odd cells for inverse-variance calibration and even cells for comparison. The fitted mean is 0.9224361066 and approximate uncertainty 0.0038483763. Covariances are unavailable; the uncertainty widths are supplied MC quantities, not asserted to be standard errors of independent sample means.

| Cell | Observed fidelity | Supplied MC width | $\lvert z\rvert$ |
|---|---|---|---|
| 2 | 0.911629 | 0.009759 | 1.0301 |
| 4 | 0.939911 | 0.006394 | 2.3417 |
| 6 | 0.923065 | 0.007581 | 0.0740 |
| 8 | 0.902836 | 0.009855 | 1.8526 |
| 10 | 0.888098 | 0.016270 | 2.0539 |

Table S3: Within-dataset even-cell holdout. All predictions use the odd-cell calibrated mean. Shared systematics and cross-cell covariance are unavailable.

The path-point calibration is separate: pair 5,6, a 270 ns detection window, 3 K cryostat, and 18 $\mu$s total optical age comprising 10 $\mu$s AFC rephasing and 8 $\mu$s spin storage. Its conditional mean is 0.9671968089 with supplied MC uncertainty 0.0051670280. Neither the window width nor a delay in an ambiguous figure is substituted for the total optical age. A query at another storage time is rejected; no $T_1$ or $T_2$ is supplied at this point.

The acceptance model multiplies 0.16 AFC efficiency, 0.42 two-way optical transfer, 0.50 demultiplexing and 0.42 temporal acceptance. The first three reported component uncertainty widths are 0.01, 0.02 and 0.04. Under independent errors and exact last factor,

$$\sigma_\eta = \eta\sqrt{(0.01/0.16)^2 + (0.02/0.42)^2 + (0.04/0.50)^2} = 0.0015824218.$$

Temporal-acceptance uncertainty, cross-component covariance, fiber/filter/detector losses and false heralds are not included. Network Monte Carlo intervals condition on the nominal acceptance.

Readout-window diagnostics calibrate on 80 widths at or below 600 ns and assess 91 widths at or above 800 ns; 40 intermediate widths are unused. Three signal-plus-background fits have maximum pointwise discrepancies 1.728745, 2.638455 and 0.722965 for pairs 1,2; 5,6; and 9,10. Their corresponding constant-fidelity discrepancies are 5.547306, 5.279383 and 4.118887. The pair 5,6 fit reaches signal-fidelity upper bound one. Window measurements share nested counts and lack covariance, so these screens are not calibrated global hypothesis tests or $T_2$ estimates.

# S4 Mean fidelity does not uniquely determine a channel

This is a constructive mathematical example, not a second fit to all released experimental states. For the same six-state mean $F_0$, define $p = 2(1 - F_0)$ and $q = 3p/4$. Depolarizing Kraus operators are

$$K_0 = \sqrt{1 - 3p/4}\, I, \quad K_x = \sqrt{p/4}\, X, \quad K_y = \sqrt{p/4}\, Y, \quad K_z = \sqrt{p/4}\, Z.$$

Phase-flip operators are $M_0 = \sqrt{1-q}I$ and $M_1 = \sqrt{q}Z$. The latter leaves the two Z-axis states unchanged and gives fidelity $1 - q$ for each of the four transverse states. Its six-state mean is $1 - 2q/3 = 1 - p/2 = F_0$.

For identical independent maps on the two halves of a Bell pair,

$$\begin{aligned} F_B^{\mathrm{dep}} &= 1 - \tfrac{3}{2}p + \tfrac{3}{4}p^2, \\ F_B^{\mathrm{phase}} &= (1-q)^2 + q^2 = 1 - \tfrac{3}{2}p + \tfrac{9}{8}p^2, \\ F_B^{\mathrm{phase}} - F_B^{\mathrm{dep}} &= \tfrac{3}{8}p^2. \end{aligned}$$

At $F_0 = 0.9671968088636844$, $p = 0.06560638227263116$ and $q = 0.04920478670447337$. Direct Kraus propagation gives $F_B^{\mathrm{dep}} = 0.9048185746372300$ and $F_B^{\mathrm{phase}} = 0.9064326486603184$, a difference of 0.0016140740230883. The matrices and values are regenerated by `scripts/make_figures.py`. More extensive tomography and a conditioning model are needed to discriminate experimentally compatible channels.

# S5 PR2026 curve diagnostics and negative outcomes

Four released classical-efficiency curves have 15, 70, 70 and 70 observations for no RF, XX, XY4 and XY16. Efficiencies and their supplied SEM quantities are converted consistently; reported total delays are converted from microseconds to seconds. The fit factor is explicitly one, defining an apparent retrieval-decay scale. No result is converted to homogeneous spin $T_2$.

The Gaussian model is $b + A\exp[-(t/\tau)^\beta]$, with exponential $\beta = 1$ or stretched $0.25 \le \beta \le 4$. The implementation uses three starts and bounded log-$\tau$. Every fourth sorted unique time group, using zero-based group indices congruent to three modulo four, is withheld. Thirty parametric bootstrap draws are requested per full-data fit. Status includes the full-data approximate Gaussian goodness diagnostic, holdout residuals, boundaries, observed-time range, reviewed observable mapping and bootstrap stability.

| Sequence | Model | $\tau$ (ms) | Held RMSE | Approx. $p$ | Status |
|---|---|---|---|---|---|
| XX | Exp. | 0.69686 | 3.582 | 5.8e-90 | Mismatch |
| XX | Stretch. | 0.0129498 | 2.853 | 1.16e-41 | Mismatch |
| XY16 | Exp. | 348.25 | 1.983 | 2.42e-87 | Mismatch |
| XY16 | Stretch. | 2.39489 | 1.417 | 8.34e-59 | Mismatch |
| XY4 | Exp. | 1.16409 | 3.786 | $< 10^{-300}$ (underflow) | Mismatch |
| XY4 | Stretch. | 0.0954074 | 3.434 | $< 10^{-300}$ (underflow) | Mismatch |
| none | Exp. | 0.0188799 | 0.622 | 0.995 | Inconclusive |
| none | Stretch. | 0.0288762 | 0.468 | 0.998 | Inconclusive |

Table S4: Apparent classical retrieval scales and diagnostics; none identifies physical spin $T_2$. Extremely small probabilities depend on the Gaussian independence model.

| Sequence | Model | $\beta$ | Requested | Successful |
|---|---|---|---|---|
| XX | Exp. | 1 | 30 | 30 |
| XX | Stretch. | 0.25 | 30 | 30 |
| XY16 | Exp. | 1 | 30 | 15 |
| XY16 | Stretch. | 3.2387 | 30 | 30 |
| XY4 | Exp. | 1 | 30 | 30 |
| XY4 | Stretch. | 0.25 | 30 | 30 |
| none | Exp. | 1 | 30 | 30 |
| none | Stretch. | 2.0866 | 30 | 28 |

Table S5: Requested and accepted parametric-bootstrap draws. Additional issue flags remain in the full JSON outputs.

All six RF fits are model mismatches. The no-RF exponential scale lies outside the observed window; the no-RF stretched bootstrap yields only 28 of 30 accepted draws and is inconclusive under the implemented requirement. A zero stored $p$ is numerical underflow, not an exact zero probability. Tiny Gaussian probabilities are conditional on supplied uncertainties and independence; unmodeled correlation/systematics limit the physical interpretation. The source experiment already reports nonexponential behavior. Thus the retained negative results diagnose the tested envelopes, rather than demonstrating that the experimental memory itself fails.

The separate heralded-photon arrays are not qubit process tomography. The source value `st`=100 $\mu$s and the article's 100.9 $\mu$s value are retained as an unresolved clock mapping. The paper's nominal spin coherence quoted from other work is not a new lifetime measured from these arrays. Neither $g^{(2)}$ nor extended classical storage is substituted for time-resolved qubit fidelity.

# S6 Deterministic and synthetic verification

The Markov case supplies $\Gamma_1 = 10$ and $\Gamma_\phi = 20\ \mathrm{s}^{-1}$ at hypothetical 4.2 K conditions, giving 40 ms. Rate-box corners use $\Gamma_1 \in [8, 12]$ and $\Gamma_\phi \in [18, 22]\ \mathrm{s}^{-1}$. Their range [35.714286,45.454545] ms is deterministic parameter uncertainty; its $-10.7143\%/+13.6364\%$ deviations are not prediction-error margins.

The finite-bath case has one $^{13}$C spin $I = 1/2$, logical projections $m = \pm 1/2$, laboratory Zeeman $\omega_z = 214163.9829$ Hz and hyperfine $A_{zz} = 4000$ Hz at 4.2 K. Other couplings vanish. In the unpolarized

limit, the longitudinal model gives $L(t) = \cos(\pi A_{zz} t)$; the finite-temperature implementation uses its exact Gibbs state. Each of free evolution and ideal one-pulse Hahn control evaluates 101 times over 0–1 ms. Free evolution has a first sampled $1/e$ crossing bracket [90,100] $\mu$s and revivals. Hahn control swaps the commuting branches and has no sampled crossing over this finite window. These parameters are specified controls, not inferred properties of a molecular candidate.

The synthetic likelihood control uses $A = 0.8$, $b = 0$, $\tau = 1$ ms, exponential factor one and four known phases $0, \pi/2, \pi, 3\pi/2$. Ages are zero plus 20 geometric points from 10 $\mu$s to 4 ms. At every age and phase, 4000 binomial trials are generated with seed 20261010: 84 count observations and 336,000 Bernoulli trials. Group holdout retains all four phases at each withheld age together. Five ages (20 observations) are held out; 64 observations calibrate the holdout fit. The reported full-data fit uses all 84 counts. Sixty requested bootstrap draws all succeed. Recovery is 0.9796631569 ms with conditional 95% interval [0.9502149602,1.0077371898] ms, held-out Pearson RMSE 0.8632951 and approximate goodness $p = 0.9528551$. One recovered generator instance is not a coverage study.

# S7 Network controls and work counts

The Teller integration control uses the actual SeQUeNCe 1.2.0 event kernel and density manager. Each attempt loads an ideal Bell state, schedules retrieval at 18 $\mu$s and applies the calibrated map once. Two nodes have nominal acceptance 0.014112, independent losses and ideal link/herald probability one. Four seeds 20261010–20261013 each have 100,000 attempts at a 1 ms interval. Counts 20, 32, 24 and 11 sum to 87. The fixed-loss expectation is 79.6594176 pairs. The pooled Wilson interval is [0.000176350196,0.000268249181]. Conditional Bell fidelity is 0.9048185746, and the maximum joint-density discrepancy is $1.11 \times 10^{-16}$. The simulated attempt clock, totaling 400 s, is not wall-clock runtime or a physical network benchmark.

The separate hypothetical profile has write/read acceptance 0.8/0.7, depolarization parameters 0.01/0.02, write/read durations 100/200 $\mu$s and constant 4.2 K stages. It exports ages 0, 100 and 500 $\mu$s; retrieval uses 500 $\mu$s, ideal link/herald factor 0.5, 2000 attempts and 1 ms interval. It produces 321 pairs and Bell fidelity 0.9559176317. Its maximum channel Choi discrepancy is $4.00 \times 10^{-15}$ and joint-density discrepancy $5.55 \times 10^{-17}$. These supplied parameters have no experimental calibration in this study.

| Executed/recorded unit | Count |
|---|---|
| Processed Teller state-fidelity values | 90 |
| Spatial held-out cells | 5 |
| Primary window fits / constant diagnostics | 3 / 3 |
| PR classical curve observations | 225 |
| Primary PR curve-model fits | 8 |
| Requested PR bootstrap replicates | 240 |
| Deterministic model specifications | 3 |
| Exact spin-bath time evaluations | 202 |
| Markov parameter-box corners | 4 |
| Synthetic count observations / Bernoulli trials | 84 / 336,000 |
| Synthetic requested bootstrap replicates | 60 |
| Teller network seeds / attempts | 4 / 400,000 |
| Hypothetical network attempts | 2000 |
| Original / additional evaluation tests passing | 75 / 7 |
| New hardware experiments / ML training runs | 0 / 0 |

Table S6: Different work units are reported separately. Optimizer function evaluations, three starting points, holdout refits and bootstrap refits are not counted as eight additional primary experiments. The additional bounded local worker benchmark is reported separately below; cluster-scale performance remains untested.

# S8 Additional system evaluation: complete executed protocol

Steps 1, 3 and 4 have executed results at their stated software scope. Step 2 has an executed synthetic second-family case, whereas its experimental physical objective remains blocked. The four selected records share evaluation-source fingerprint

`d8fc6d2ecb580e0ba1c8881ffbd542d087c493b7621cb89509948e595211ec92`.

Their common lab-source fingerprint is

`9c4490acf2e5a42e8e10007989eeb94b5e406b17f5d187ca310847581ee2edb7`.

These fingerprints identify source content, not hardware or an external scientific endorsement. Steps 2 and 3 were confirmed with the published implementation after their initial runs; the initial outputs remain archived. Table generation asserts the shared fingerprint and expected numerical/status gates without overwriting the original run records. The configuration digest and UTC timestamp of each selected record are in `evidence/system_evaluation_claim_map.json`. Generated table files are under `evaluation/paper/`.

## S8.1 Step 1: independent reference equations and matched inputs

The reference `evaluation/wqeval/reference.py` imports NumPy/SciPy without importing `memorylab`. For column vectorization and Hamiltonian $G = \mathrm{diag}(0, \delta)$ in Hz, it uses

$$\mathcal{L} = -2\pi i(I \otimes G - G^{\mathsf{T}} \otimes I) + \sum_J \left[J^* \otimes J - \tfrac{1}{2} I \otimes J^\dagger J - \tfrac{1}{2}(J^\dagger J)^{\mathsf{T}} \otimes I\right].$$

The jump operators are $\sqrt{\Gamma_1(1-p_e)}\,|0\rangle\langle 1|$, $\sqrt{\Gamma_1 p_e}\,|1\rangle\langle 0|$, and $\sqrt{\Gamma_\phi/2}\,Z$. The stage superoperator is $\exp(t\mathcal{L})$. Write and read stages include their specified depolarizing maps, and storage adds the declared network-activity dephasing. The reference constructs $C = \sum_{a,b} |a\rangle\langle b| \otimes \mathcal{E}(|a\rangle\langle b|)$ by matrix-unit propagation, including an orthogonal erasure outcome for the full channel. It computes six Pauli-eigenstate outputs and their mean target fidelity independently of the lab's Kraus construction.

The four joint parameter choices $(\Gamma_1, \delta, p_e)$ are $(0, 0, 0)$, $(0.05, 123, 0.1)$, $(10, -500, 0.5)$ and $(100, 1000, 0.9)$, with rates in $\mathrm{s}^{-1}$ and detunings in Hz. Each is combined with storage temperatures 4.2, 10 and 80 K and ages $[0, 10^{-6}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}]$ s. Write/read stages remain at 4.2 K, last $10^{-4}/2 \times 10^{-4}$ s and use acceptance 0.8/0.7 and depolarization 0.01/0.02. The pure-dephasing rate is $0.01 + 0.001T + 0.001(T/10)^3$ $\mathrm{s}^{-1}$, erasure rate 0.01 $\mathrm{s}^{-1}$, and storage activity contributes $10 \times 0.002 = 0.02$ $\mathrm{s}^{-1}$. Rate validity is declared over 1–300 K. These are chosen software profiles, not fitted material parameters. The measured Teller point supplies the thirteenth profile and its sole supported age of 18 $\mu$s at 3 K, giving $12 \times 6 + 1 = 73$ channel-age points.

The two experimental input files are `4.csv` and `8a.csv` from record 14929997:v1. Their SHA-256 hashes, original download URLs and data-license metadata are pinned in `evaluation/data/teller_sources.json`. Both parsers use the same checked bytes. The path file yields 30 values and mean 0.9510824228990328; the time-bin file yields 60 values and mean 0.9140479475068066. The discrepancy is zero for every processed value. The comparison threshold is $10^{-8}$ for the full Choi matrix, six states, average fidelity, acceptance and processed values. This threshold is numerical, not an experimental accuracy margin.

Both timing arms are warmed up before five repeats, with their order randomized by seed 20261010. All 73 channel constructions are included in each repeat. `construction_timing.csv` retains every measured duration and actual order. The arms do not perform equal amounts of metadata, export, validation or metric-construction work. File IO, source retrieval, network events, fits and electronic structure are excluded. Native thread environment variables are recorded but were not constrained for step 1. The medians cannot be interpreted as a fair end-to-end speedup or engineering-effort reduction. `related_tools_audit.json` distinguishes executed comparison from documentary context; AiiDA, PySCF, PyCCE, QuTiP and SeQUeNCe were not separately timed here. Existing AiiDA provenance and caching are not presented as new contributions.

## S8.2 Step 2: synthetic control, curve gate and physical stop

The model/curve comparison requires the observable `complex_pure_dephasing_coherence`, explicit exclusion of population relaxation, source hashes, curve reviewer, normalization and observable mapping. A measured case additionally requires reviewed model-parameter provenance, a nonhypothetical model and exact equality of full declared experimental conditions between curve and model. The fixed-model runner

accepts `exact_spin_bath` or optional `pycce`; it fits zero parameters. The initial curve value must be unity, magnitudes must lie in the unit disk and a strictly increasing finite grid must start at zero. No silent clipping of noisy or unnormalized measurements is provided.

The executed control has one $^{13}$C nuclear spin $I = 1/2$, central logical projections $\pm 1/2$, laboratory $A_{zz} = 4000$ Hz and $\nu_z = 214163.9829$ Hz, zero other couplings, 4.2 K and zero pulses. Forty-one equally spaced ages span 0–1 ms. An independent closed-form Gibbs-initialized FID produces the curve; the finite Hamiltonian solver supplies the comparison. Every complex residual is divided by the chosen scale 0.01, and the maximum must be at most three. The observed maximum is $3.7192473 \times 10^{-12}$. This chosen scale is not an experimental SEM, and the gate has no calibrated statistical false-positive rate. The 75–100 $\mu$s crossing bracket differs from the earlier 101-point control's 90–100 $\mu$s bracket solely because the grids differ; both are sampled numerical brackets and both detect revivals.

All write/store/read temperatures are hypothetically 4.2 K. The supplied interface has acceptance 0.8/0.7, depolarization 0.01/0.02 and durations 100/200 $\mu$s, with the independent rate models retained from the example. Export requires `-allow-hypothetical`. At 500 $\mu$s, the provider-neutral two-node workload has 10,000 trials, herald probability 0.1, 1 ms interval, Bell-fidelity threshold 0.8 and seed 20261010. It yields 1020 heralds and 313 retrieved/quality-qualified pairs. Conditional Bell fidelity is 0.9559176317211326; the retrieved-fraction Wilson interval is [0.0280630600,0.0348969001]. The simulated qualified rate is 31.3 $s^{-1}$ under these supplied assumptions, not measured network performance. This workload is not the earlier 2000-trial SeQUeNCe control.

The separate pending descriptor targets a molecular reference at DOI 10.1039/C8SC04122A. Running it returns `blocked_missing_reviewed_data` and exit code two, with no physical validation and no network export. Required inputs are a reviewed normalized curve with measurement uncertainties, fixed Hamiltonian and tensor frames, field/temperature/isotopes/pulses, separation of relaxation from pure dephasing, and independent optical-interface review. Reported molecular phase-memory time alone does not satisfy these requirements. Thus no second-family experimental reproduction or new-material lifetime is supplied by step 2.

## S8.3 Step 3: individual fault interventions

The suite creates local copies of inputs and explicitly unsafe bypasses without modifying production guards. Each invalid case has a valid positive control that must succeed. Rejection alone would be insufficient if every valid input also failed. All seven positive controls and all seven guarded rejections pass; the deliberate bypasses accept all seven cases.

| Constructed fault | Guarded result | Accepted unsafe-bypass consequence |
|---|---|---|
| Operating-point age extrapolation | Blocked | 10× unsupported age |
| Operating-point temperature extrapolation | Blocked | 1 K unsupported shift |
| Total coherence rate reused as pure dephasing | Blocked | 16.667% conditional $T_2$ error |
| Unreviewed noise independence | Blocked | One unsupported declaration |
| Hypothetical channel without opt-in | Blocked | Two hypothetical age samples |
| Absolute-age map applied incrementally | Blocked | 0.097247 maximum element error |
| Modified source bytes | Blocked | 1000× forged source value |

Table S7: Step 3: all seven valid positive controls pass; all invalid cases are rejected. Consequences have distinct units and are not aggregated into an accuracy score.

The age bypass changes the declared sole point age before evaluating a tenfold longer age, thereby illustrating an unsupported extrapolation rather than measuring its physical error. The temperature bypass changes the declared point temperature by 1 K, again with no experimentally known extrapolation error. Noise independence is counted as an unsupported declaration; evidence promotion counts two age samples. The rate-definition example has $\Gamma_1 = 10$ $s^{-1}$ and correct independent $\Gamma_\phi = 20$ $s^{-1}$, giving 40 ms. Treating total coherence rate 25 $s^{-1}$ as pure dephasing gives 33.333 ms instead. The modified-source case changes a chosen value from 0.001 to 1 s; it exercises the new evaluation source contract, whereas the other six cases exercise existing lab guards.

In the incremental-application control, the 1 ms complete channel includes write/read depolarization as well as storage; applying it twice produces maximum density-element discrepancy 0.0972467329 relative to

applying it once. A separate units control converts 1 ms correctly to 0.001 s and contrasts it with an omitted conversion to 1 s; its maximum element discrepancy is 0.1322008762. These consequences have different units and scientific meanings. Neither their average nor seven out of seven represents a general material-prediction accuracy or real-world researcher error rate. Equivalent guards may be implemented in other frameworks.

## S8.4 Step 4: workload, timing scope and robustness

The native runner workload has Markov and one-spin exact-bath models, detunings $[0, 123, 500]$ Hz, two replicates and six ages $[0, 10^{-4}, 5 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}, 5 \times 10^{-3}]$ s. Replicates have distinct deterministic seed streams; they are not independent hardware measurements or stochastic bath realizations in this deterministic workload. The Cartesian count is $2 \times 3 \times 2 = 12$ jobs per study. Master seed is 20261010. Three repeats independently randomize worker order among 1, 2 and 4. Worker count is excluded from physics identity, allowing a direct consistency check across scheduling choices.

Each cold study runs in a separate output directory and an isolated measurement process. OMP, MKL, OpenBLAS and NumExpr thread limits are one in those benchmark subprocesses. Per-job timeout is 60 s, driver timeout 600 s and max-run budget 24. The measured interval includes process startup, actual channel construction and artifact IO. The warm interval reuses terminal-result checkpoints; it is not electronic-structure memoization. Nine cold studies complete 108 jobs, and nine warm passes reuse 108 results. The scientific digest, excluding durations and output paths but including configuration identities and channel data, is identical across all nine studies.

| Workers | Cold median [min,max] (s) | Resume median (s) | Median jobs/s |
|---|---|---|---|
| 1 | 3.8674 [3.8089, 3.9101] | 0.01814 | 3.103 |
| 2 | 1.9781 [1.9165, 2.0665] | 0.01741 | 6.066 |
| 4 | 1.1782 [1.1299, 1.2286] | 0.01817 | 10.185 |

Table S8: Step 4: three local cold studies and three checkpoint-reuse passes per worker count, with 12 jobs per study. Bounds are observed minima/maxima, not confidence intervals.

The reported bounds are observed ranges from three repeats, not confidence intervals. Separate resource measurements use process usage counters: parent maximum RSS is 34.578125–37.078125 MiB and the largest terminated child's maximum is 63.238281–66.414063 MiB. The metric is not peak total memory across concurrent workers. Child CPU seconds and each measurement's raw counters remain in the retained summary. Performance applies to this Linux environment and small supplied models, not general CCE, electronic structure, all elements or cluster-scale workloads.

Three additional robustness probes are distinct from the 108 cold jobs. A profile without write temperature produces `missing_data`. A real worker process with a $10^{-6}$ s timeout produces `timeout`; this deadline intentionally tests termination and is not a solve-time estimate. A precompleted valid job followed by the full study completes all 12 jobs with exactly one checkpoint reused. Thus robustness adds one missing-data worker, one timeout worker and 12 successful calculations across the partial/restarted study. No forced-kill during an atomic write, power-loss, corruption or concurrent cache-writer experiment was performed. Failed terminal checkpoints are preserved as failures, not retried or counted as successful physics results.

## S8.5 Core fresh-checkout CI and traceable work units

GitHub Actions run 38051481969 at commit `de52b32f0c34a05cf4f6a0c01bca6ee087e6cb67` uses Ubuntu 24.04 and Python 3.12.15. It installs the core editable package with NumPy 2.5.3 and SciPy 1.18.1, passes seven evaluation tests, automatically retrieves/checks both pinned experimental files, and reruns the matched comparison. Its maximum full-Choi discrepancy is $1.3214015 \times 10^{-14}$ versus the $10^{-8}$ gate. `evidence/core_ci_job.log` and `core_ci_verification.json` preserve the log and summary; the log hash is checked. This core CI result is additional environment evidence, not full optional-backend reproduction or colleague-led independent scientific review. CSV bytes are preserved and automatic Git line-ending conversion is disabled to protect source hashes.

The original 75 tests plus seven additional evaluation tests constitute 82 distinct local passing tests. Repeating controls, CI tests, numerical ages or Monte Carlo trials does not increase that count. Work units remain separate: 73 channel-age comparisons, 90 processed data values, 41 synthetic coherence ages, 10,000 provider-neutral network trials, seven selected invalid contracts with seven positive controls, 108 cold worker

jobs and 108 checkpoint reuses. No new hardware experiment or ML training run is counted. Steps 5 and 6 remain the author's and colleagues' full independent reproduction and scientific/release review tasks.

## S9 Reproduction and claim traceability

The source archive used for this study has SHA-256

`c64b94a573def9d999b65be7201947e9c`
`2ce4e496aaf201650d7c1e3d44733aa`.

The code and article are now public at https://github.com/VesterlundCoder/memory-discover-lab. The core CI release commit is recorded above; the original private source-archive hash remains a separate provenance identifier. No software DOI or peer-review acceptance is asserted. The repository root contains `paper.tex`, `supplement/supplement.tex`, `figures/`, `tables/`, `evidence/`, `scripts/` and evaluated source in `lab/`. `REPRODUCIBILITY.md` supplies dependency and LaTeX setup. A public clone requires first-run online retrieval with `python scripts/reproduce.py -online`; the original private bundle or an already populated cache can use `python scripts/reproduce.py`. The script invokes, from `lab/`, the experimental baseline command, the offline deterministic command with SeQUeNCe enabled, and all unit tests, then refreshes evidence, figures and tables and checks the headline invariants. An online variant explicitly refreshes remote data and may change metadata bytes; the archived offline path reproduces the pinned study.

`scripts/verify_results.py` checks the reanalysis values, count controls, model statuses and simulated counts, and emits a claim-to-result JSON map. These checks protect the reporting pipeline; they are not extra experimental measurements. Every headline claim links to an executed result or log. Detailed fit outputs retain status, seed, bootstrap success, conditions and limitations. Files predating this article remain under `lab/` as historical source outputs and are not all fresh experimental validations. The `evidence/` directory is the article's selected result snapshot.

The system-paper tables and claim map are regenerated with `python scripts/document_system_evaluation.py`; the numerical result files are left intact. The complete manuscript can be built using `make pdf`. Numerical reproduction does not require LaTeX; PDF compilation does not require running the simulator. Figures require Matplotlib. No ML libraries are needed. The software test logs record Python 3.12.14, NumPy 2.5.3, SciPy 1.18.1, QuTiP 5.3.1, PySCF 2.14.0, PyCCE 1.1.1 and SeQUeNCe 1.2.0. Optional-adapter test execution does not replace application-specific physical validation.

The four evaluation commands use fresh `-out` directories: `python scripts/evaluate_system.py step1`, `step3` and `step4`, plus `step2 -config evaluation/data/software_control/case.json -allow-hypothetical`. The pending physical descriptor is the default for `step2` and returns exit two. Full commands and source contracts are in `evaluation/README.md`; manuscript-ready evaluation prose is under `evaluation/paper/`.

Future candidate evaluation should retain these distinctions: independently review units, definitions and conditions; anchor Hamiltonian and noise parameters; establish numerical convergence; reproduce time-resolved optical fidelity, success and noise; and assess any proposed improvement under matched operating conditions. The present study supplies the workflow and controls for such work without claiming those future outcomes.